

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture 2a: Events and Set Operations

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College of Engineering

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Outline

- ① Elements of set theory
- ② Set operations and properties
- ③ Events
- ④ Outlook
- ⑤ Appendix: De Morgan's Rule

Module 2: Probability

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Key goals for this module:

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- Understand basic set theory and operations

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- Understand basic set theory and operations
- Understand introductory probability theory
- Learn the fundamentals of conditional probability and Bayes' theorem

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- Understand events and sample spaces
- Use set theory to express combinations of events
- Understand the concepts of mutual exclusivity and collective exhaustivity of events

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The symbol “ $\subset$ ” means “is a subset of”

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$$(A \cup B) \cup C = A \cup (B \cup C) \quad (10)$$

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③ If the sample space is given by the integers in the interval $[0, 10]$, find $(A \cup B)^c$:

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Venn diagrams

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An approach for visualizing sets (sample spaces) and analyzing events.

Venn diagrams

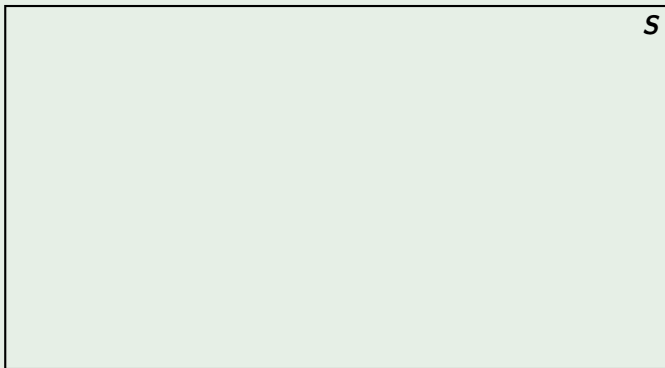
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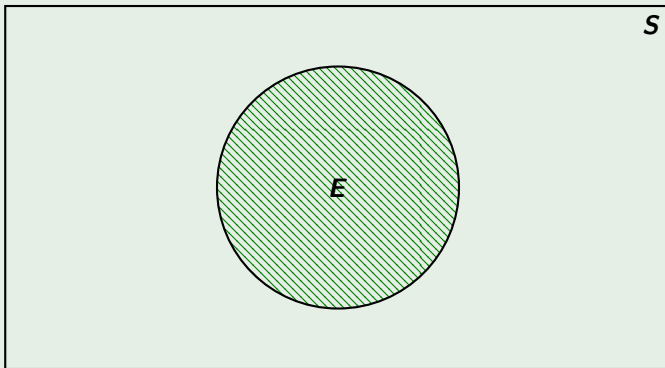
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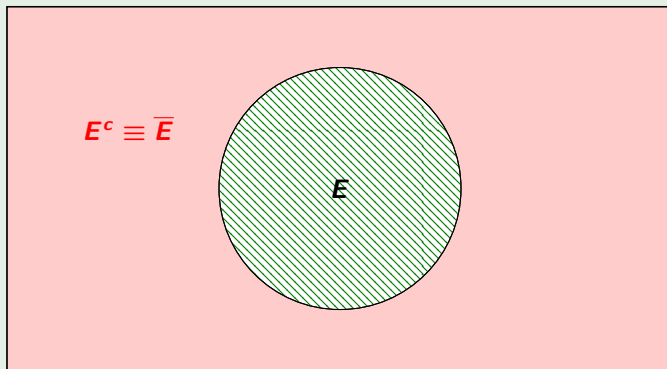
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Activity: Pizza Preference Survey

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- B = students who like mushrooms
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Activity

- Stand and sort yourselves physically into regions of the room
- Start with just sets A and B , creating a human Venn diagram
- Add set C and watch the complexity emerge
- Count each region and calculate: $|A \cup B|$, $|A \cap B|$, $|A^c|$, $|B^c|$, $|C^c|$.

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$$\bar{E} = S \setminus E \quad (17)$$

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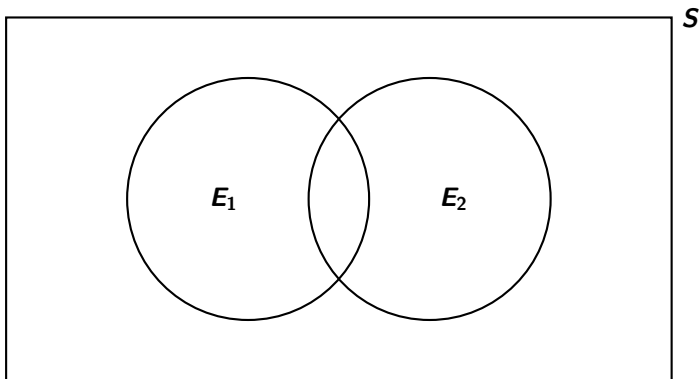
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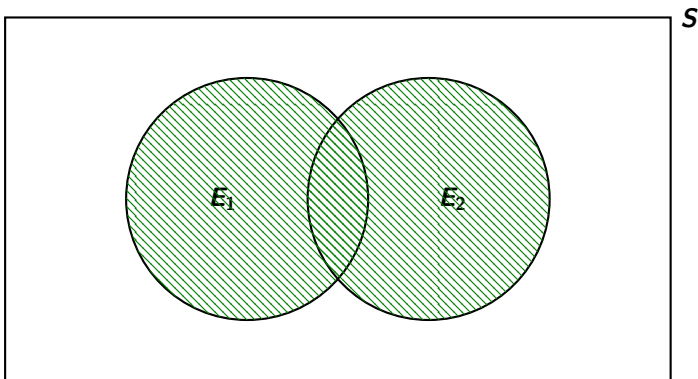
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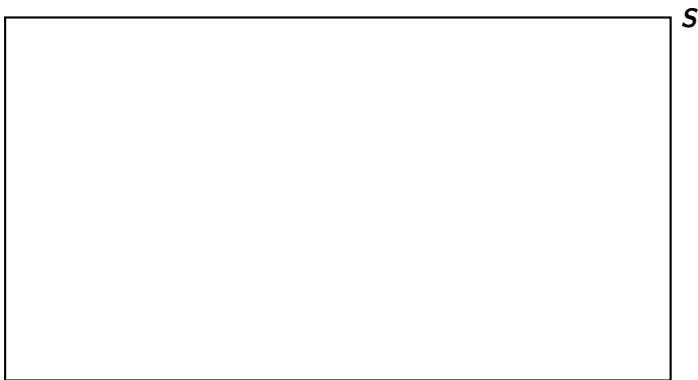
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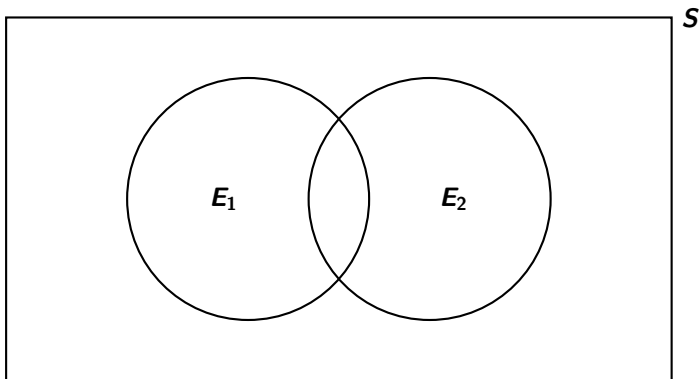
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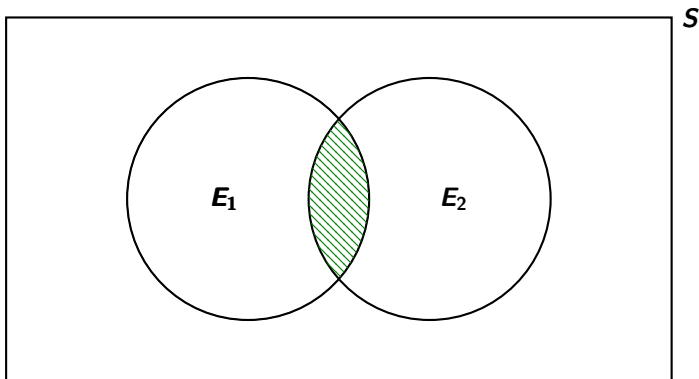
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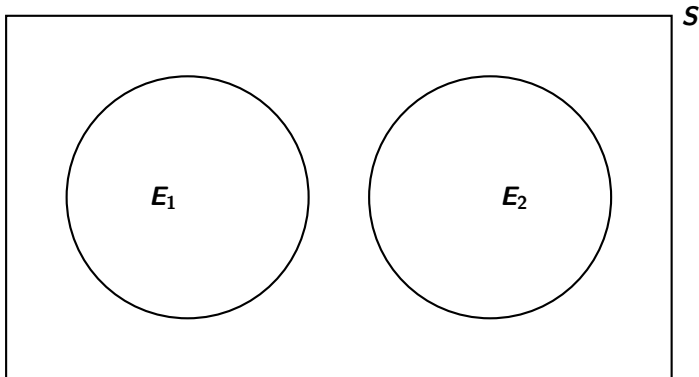


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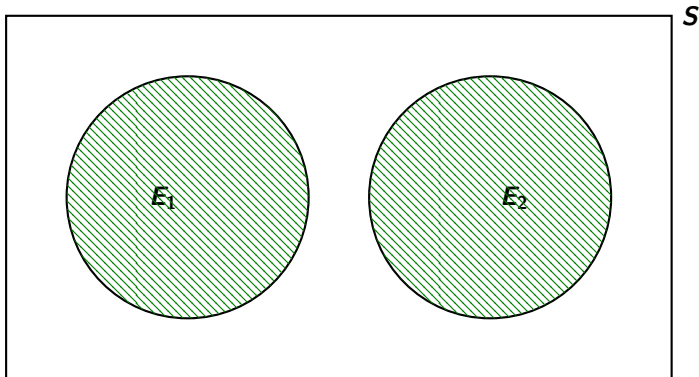


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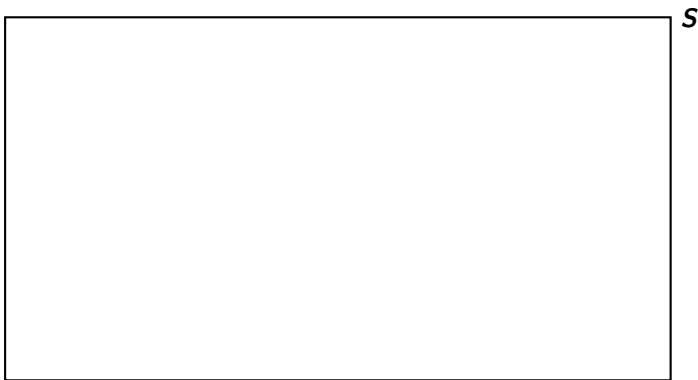
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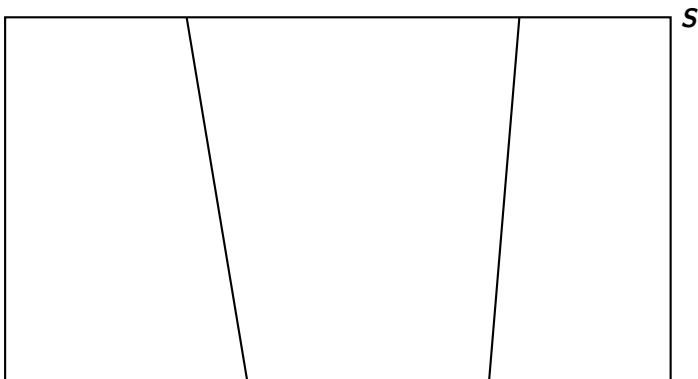
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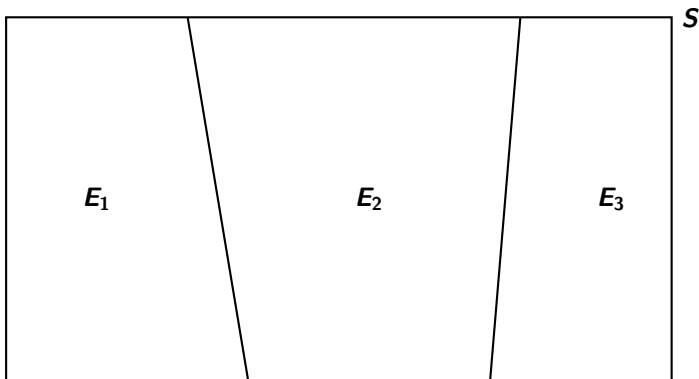
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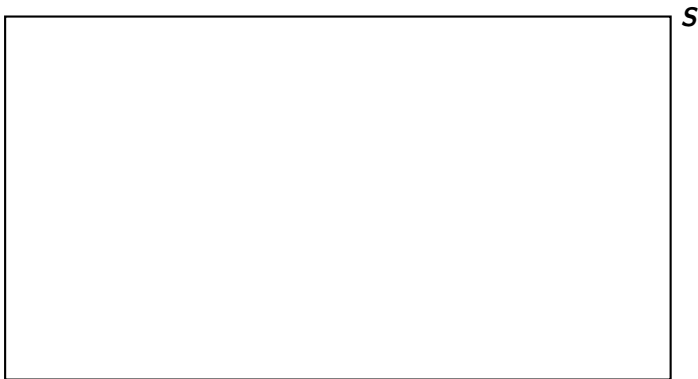
- (i) Company a submitting a bid for one project and Company b submitting a bid for another project
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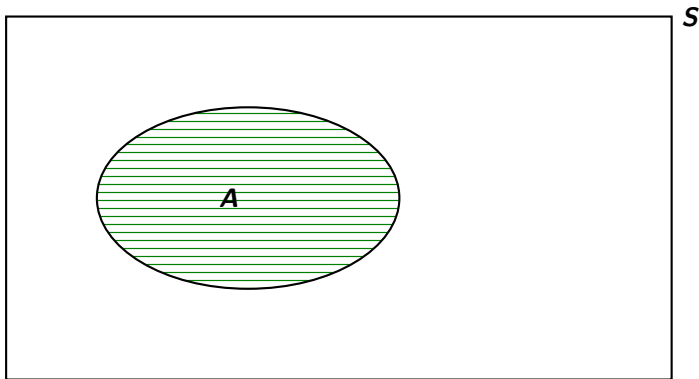
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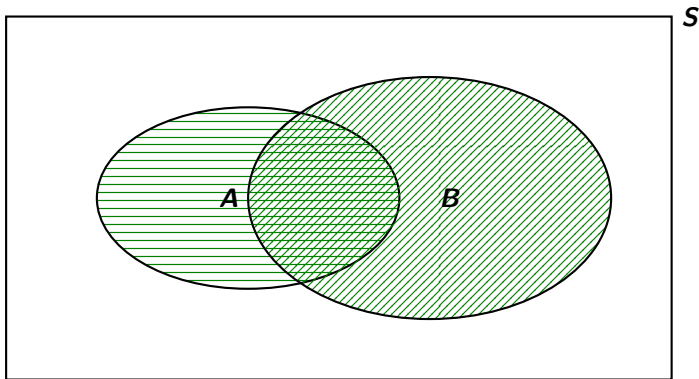
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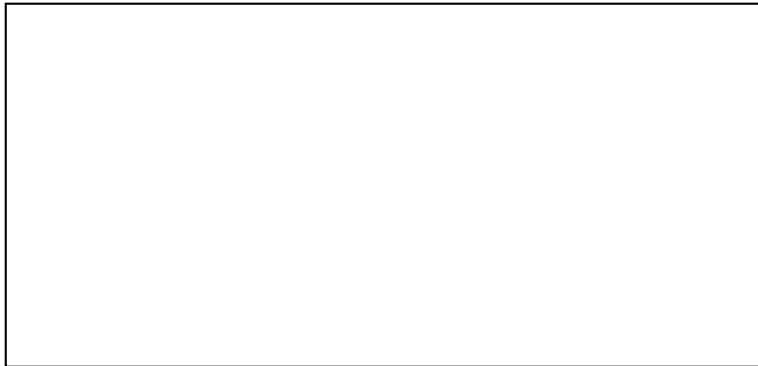
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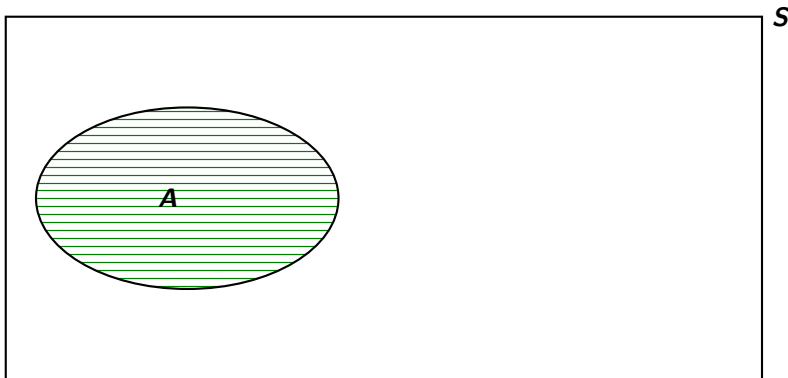
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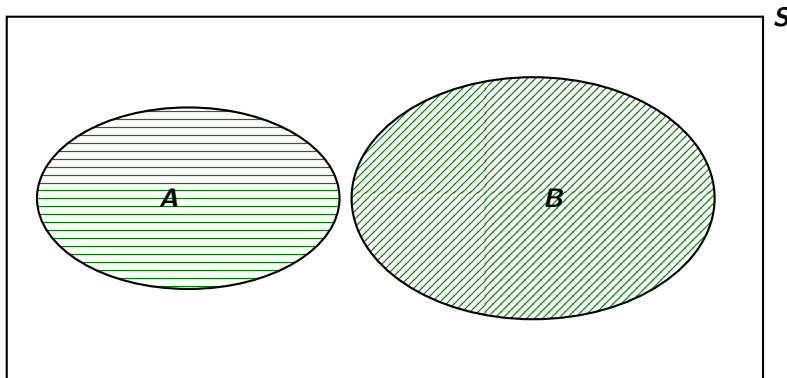
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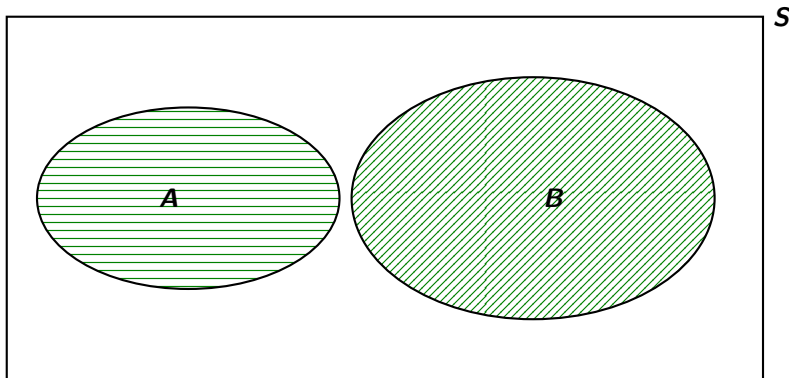
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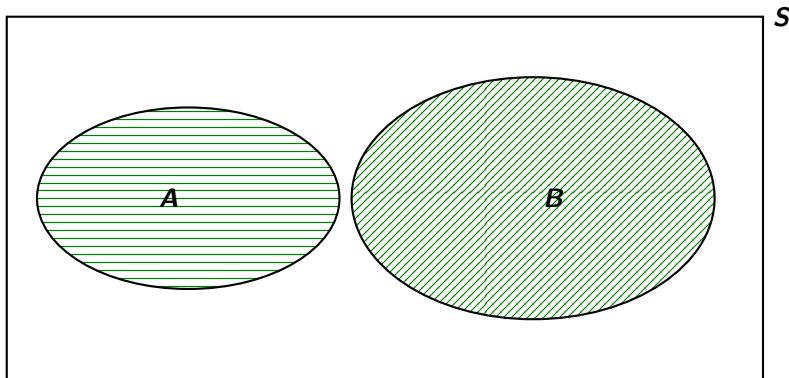
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These events are mutually exclusive, as both companies cannot win the same project.

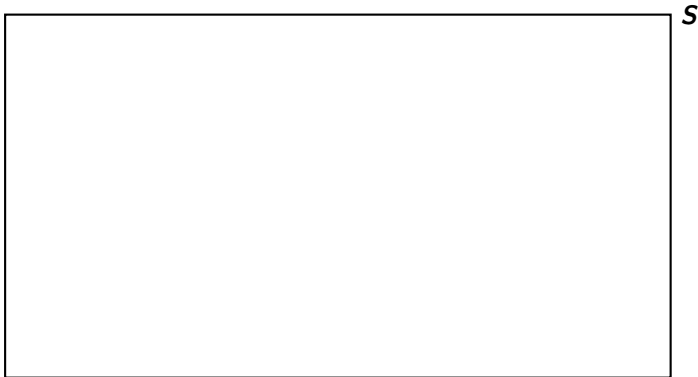
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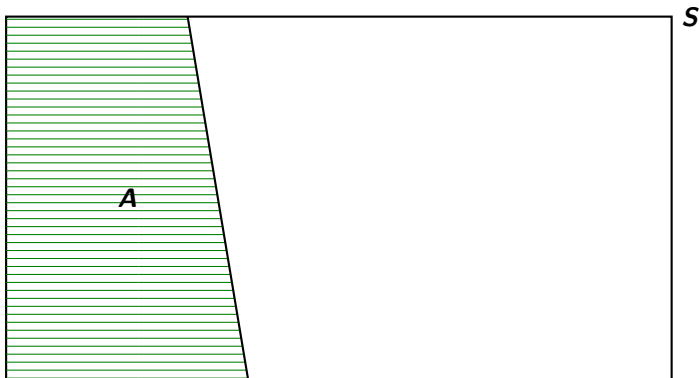
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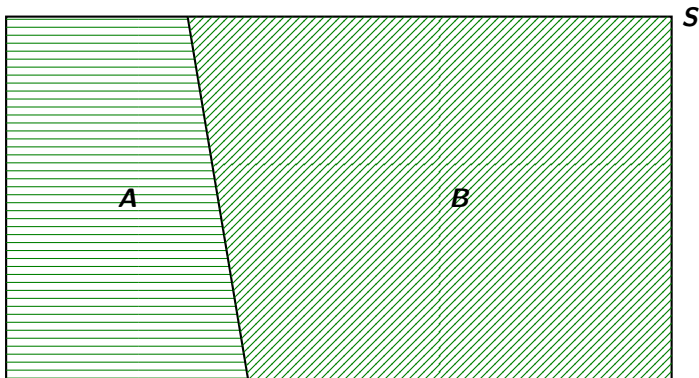
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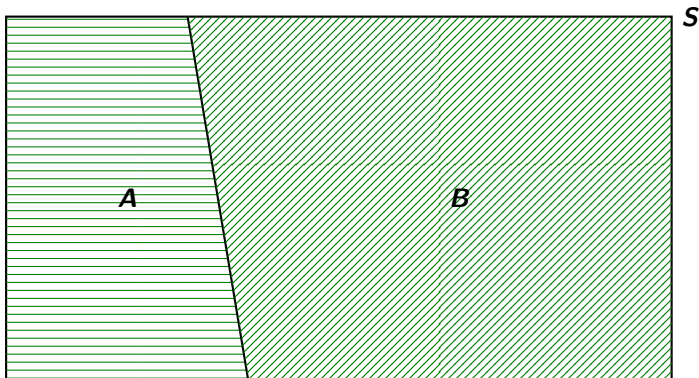
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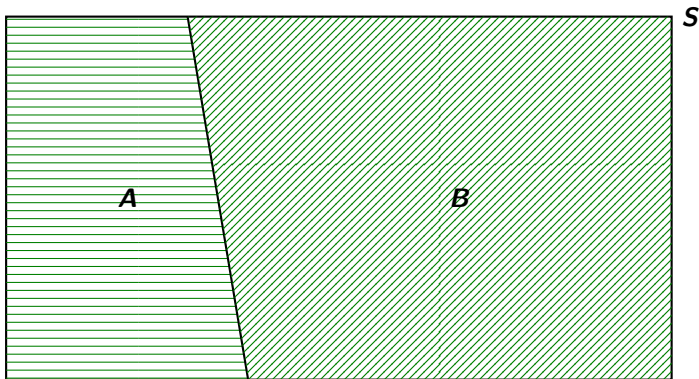
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These events are both mutually exclusive and collectively exhaustive.

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Play around with set operations: <https://seeing-theory.brown.edu/compound-probability/index.html#section1>

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Equivalently:

Complement of an intersection

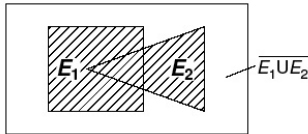
The complement of the intersection of a given number of sets/events is the union of their complements:

$$\overline{A \cap B} = \overline{AB} = \bar{A} \cup \bar{B} \quad (22)$$

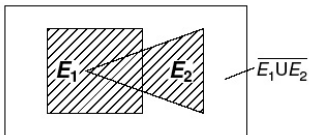
$$\overline{A \cap B \cap C} = \overline{ABC} = \bar{A} \cup \bar{B} \cup \bar{C} \quad (23)$$

$$\overline{E_1 \cap E_2 \cap \dots \cap E_n} = \bar{E}_1 \cup \bar{E}_2 \cup \dots \cup \bar{E}_n \quad (24)$$

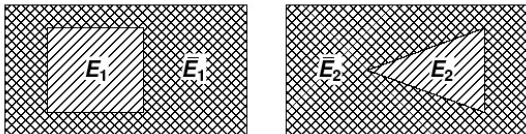
Venn diagram demonstrating de Morgan's rule



Venn diagram demonstrating de Morgan's rule

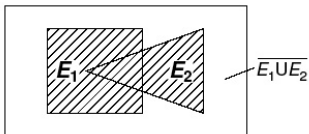


(a)

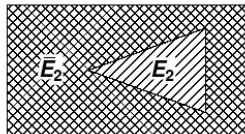
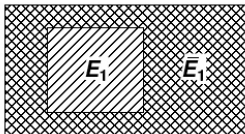


(b)

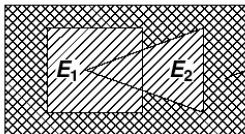
Venn diagram demonstrating de Morgan's rule



(a)



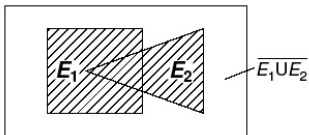
(b)



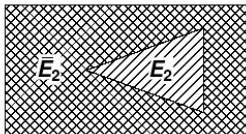
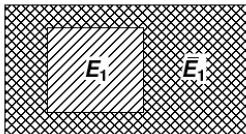
Double-hatched
region = $E_1 \cap E_2$

(c)

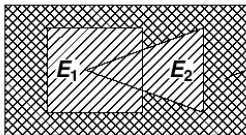
Venn diagram demonstrating de Morgan's rule



(a)



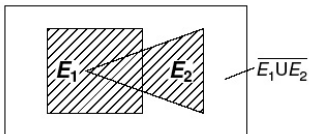
(b)



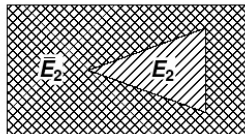
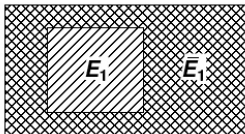
Double-hatched
region = $E_1 \cap \bar{E}_2$

(c)

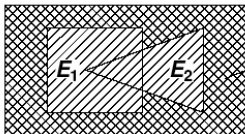
Venn diagram demonstrating de Morgan's rule



(a)



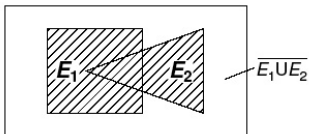
(b)



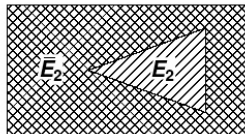
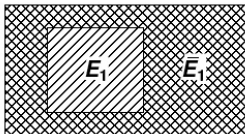
(c)

$$\overline{\overline{E_1} \cup \overline{E_2}} =$$

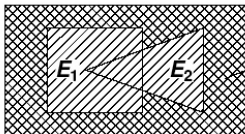
Venn diagram demonstrating de Morgan's rule



(a)



(b)



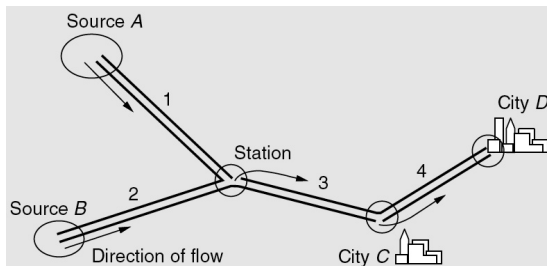
(c)

$$\overline{E_1 \cup E_2} = \overline{E_1} \cap \overline{E_2}$$

Example 4: Water supply

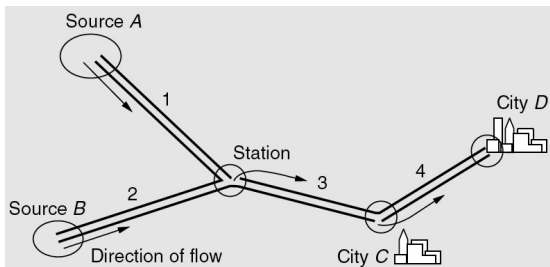
Example 4: Water supply

The water supply for two cities C and D comes from the two sources A and B . Water is transported by pipelines 1, 2, 3 and 4. Assume that either one of the two sources by itself is sufficient to supply the water for both cities. Also, denote E_1, E_2, E_3, E_4 as the failure of branches 1, 2, 3 and 4, respectively.



Example 4: Water supply

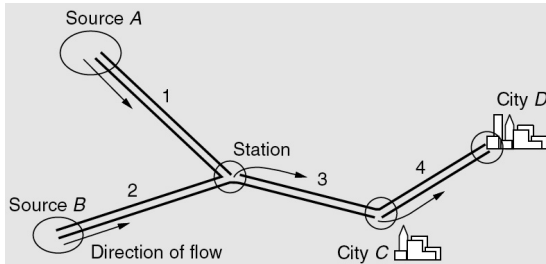
The water supply for two cities C and D comes from the two sources A and B . Water is transported by pipelines 1, 2, 3 and 4. Assume that either one of the two sources by itself is sufficient to supply the water for both cities. Also, denote E_1, E_2, E_3, E_4 as the failure of branches 1, 2, 3 and 4, respectively.



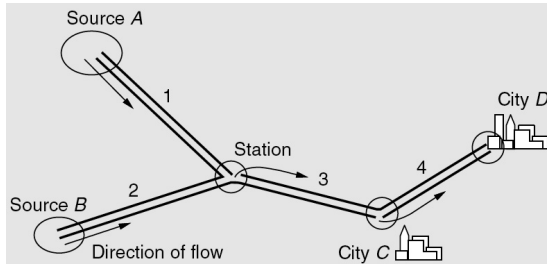
- (a) Denote the event that there is no shortage of water in C .
- (b) Denote the event that there is no shortage of water in D .

Simplify your answers using De Morgan's rule.

Example 4: Water supply (cont.)

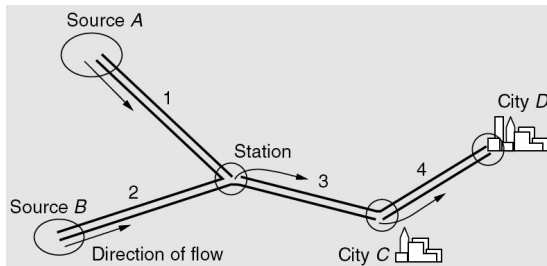


Example 4: Water supply (cont.)



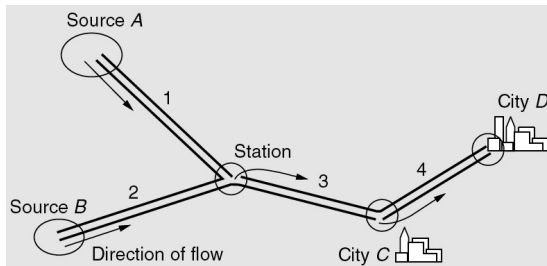
Shortage of water in C is represented by

Example 4: Water supply (cont.)



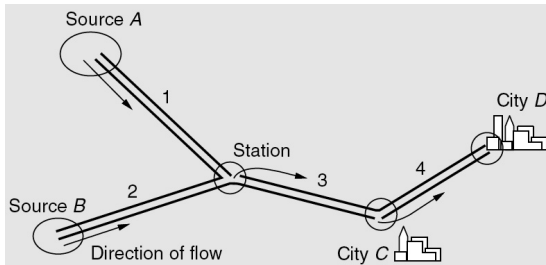
Shortage of water in C is represented by $E_1 \cap E_2 \cup E_3$.

Example 4: Water supply (cont.)



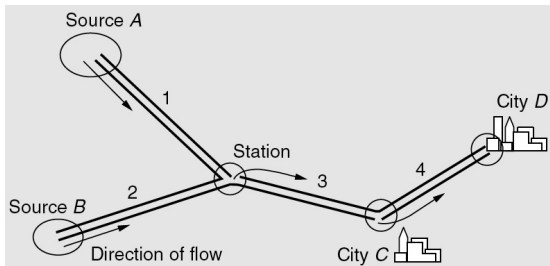
Shortage of water in C is represented by $E_1 \cap E_2 \cup E_3$. Its complement $\overline{E_1 E_2 \cup E_3}$ means there is no shortage of water in C .

Example 4: Water supply (cont.)



Shortage of water in C is represented by $E_1 \cap E_2 \cup E_3$. Its complement $\overline{E_1 E_2 \cup E_3}$ means there is no shortage of water in C . Applying de Morgan's rule, we have:

Example 4: Water supply (cont.)

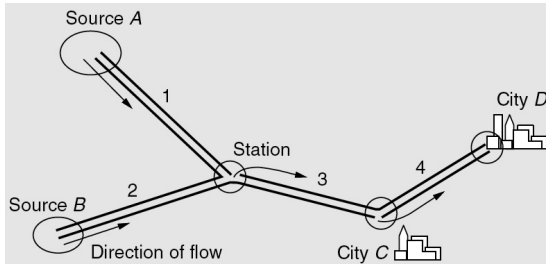


Shortage of water in C is represented by $E_1 \cap E_2 \cup E_3$. Its complement $\overline{E_1 E_2 \cup E_3}$ means there is no shortage of water in C . Applying de Morgan's rule, we have:

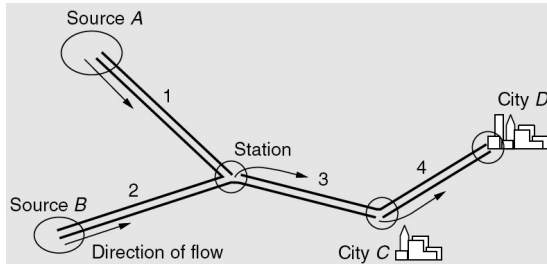
$$\overline{E_1 E_2 \cup E_3} = \overline{E_1 E_2} \cap \overline{E_3} = (\overline{E_1} \cup \overline{E_2}) \overline{E_3}$$

Example 4: Water supply (cont.)

Example 4: Water supply (cont.)

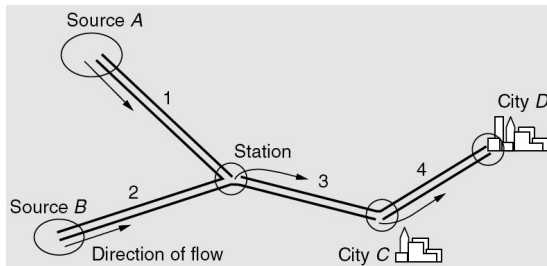


Example 4: Water supply (cont.)



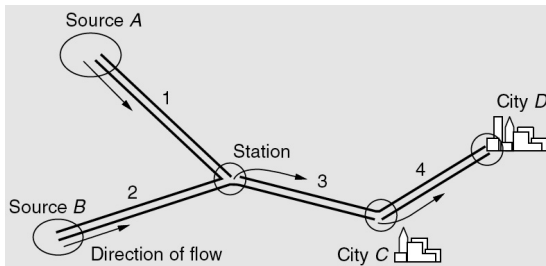
No shortage of water in D is represented by

Example 4: Water supply (cont.)



No shortage of water in D is represented by $\overline{E_1 E_2 \cup E_3 \cup E_4}$.

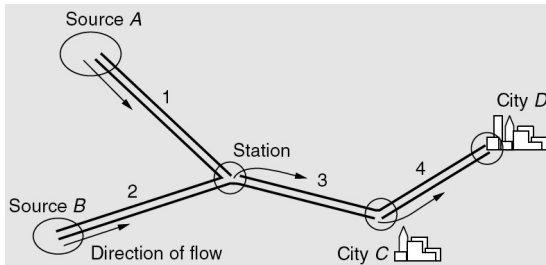
Example 4: Water supply (cont.)



No shortage of water in D is represented by $\overline{E_1 E_2 \cup E_3 \cup E_4}$.

Simplified using de Morgan's rule, this becomes

Example 4: Water supply (cont.)



No shortage of water in D is represented by $\overline{E_1 E_2 \cup E_3 \cup E_4}$.

Simplified using de Morgan's rule, this becomes $(\overline{E_1} \cup \overline{E_2}) \overline{E_3} \overline{E_4}$